Horror$Place VM-Constellation Deep Audit Report

Audit Date: 2026-04-21

Classification: Internal — Governance Review

Scope: All 13 repositories (3 public inspected; 10 private — access restricted)

Status: **PARTIAL** — Private repos require manual follow-up

Auditor: Copilot — automated schema/code/CI analysis

Requested By: Doctor0Evil

Executive Summary

This audit examines the Horror$Place VM-constellation across its full 13-repository topology, focusing on schema determinism, governance module wiring, authoring contract enforcement, and telemetry/safety data flows. The three public Tier-1 repositories (`Horror.Place`, `Horror.Place-Orchestrator`, `HorrorPlace-Constellation-Contracts`) were inspected at the file level. Ten private Tier-2/Tier-3 repositories could not be accessed and are flagged for manual follow-up.

Overall Verdict: The constellation's *schema architecture* is well-designed and

substantially complete — canonical schemas exist for spine indexing, consent state machines, safe authoring contracts, safety adapters, artifact descriptors, and repo manifests. However, the *wiring layer* between schemas and runtime code has significant gaps: no populated spine-index instance, incomplete schema-to-registry routing, missing top-level H.API modules for most governance functions, stubbed cryptographic verification, and absent enforcement of the "≤3 artifacts, one schema family" rule. The system is approximately **65% compiler-driven** and requires focused remediation to reach full audit-readiness.

4

P0 — Blocking

6

P1 — High

5

P2 — Medium

3

P3 — Low / Info

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1. Repository Topology & Access

Repository	Visibility	Tier	Audited	Core Role
`Horror.Place`	**Public**	**T1**	**YES**	Sovereign core: schemas, invariants, style/metric contracts, doctrine
`Horror.Place-Orchestrator`	**Public**	**T1**	**YES**	Cross-repo orchestrator and registry updater
`HorrorPlace-Constellation-Contracts`	**Public**	**T1**	**YES**	Contract-only: schemas, registries, CI patterns
`HorrorPlace-Codebase-of-Death`	**Private**	**T2**	**NO**	Full AI personas and Surprise.Events! logic vault
`HorrorPlace-Black-Archivum`	**Private**	**T2**	**NO**	Structured historical trauma and site invariant

Repository	Visibility	Tier	Audited	Core Role
				archives
`HorrorPlace-Spectral-Foundry`	**Private**	**T2**	**NO**	Telemetry-tuned spectral entities and SHCI personas
`HorrorPlace-Atrocity-Seeds`	**Private**	**T2**	**NO**	PCG seeds and invariant-bound region/event contracts
`HorrorPlace-Obscura-Nexus`	**Private**	**T2**	**NO**	Experimental style contracts and DSL variants
`HorrorPlace-Liminal-Continuum`	**Private**	**T2**	**NO**	Agent-sharing and cryptographic validation layer
`HorrorPlace-Process-Gods-Research`	**Private**	**T3**	**NO**	God-like personality evolution and SHCI≈0.95 research
`HorrorPlace-Redacted-Chronicles`	**Private**	**T3**	**NO**	BCI/fMRI logs and physiological horror telemetry
`HorrorPlace-Neural-Resonance-Lab`	**Private**	**T3**	**NO**	Neural resonance and haptic feedback tuning
`HorrorPlace-Dead-Ledger-Network`	**Private**	**T3**	**NO**	ZKP/ledger network, age-gating, and trust fabric

> Note:
>
> All findings below are scoped to the 3 audited public repos. Private repo compliance is inferred from contract presence and must be verified independently.

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2. Audit Area 1 — Manifests, Spine Index & Routing Determinism

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2.1 Spine Index Schema

LOCATION

`HorrorPlace-Constellation-Contracts/hpc-spine-index.schema.json`

SCHEMA ID

`https://horror.place/schemas/hpc-spine-index.schema.json`

STATUS

SCHEMA EXISTS — **NO POPULATED INSTANCE FOUND**

The spine index schema is comprehensive and well-structured. It defines:

- `schemaSpineVersion` — Semantic version of the spine document
- `updatedAt` — ISO 8601 timestamp
- `canonicalAuthority` — References to schemaRepo, invariantsSchema, metricsSchema
- `invariants[]` — Array of invariantEntry objects with: code, name, canonicalSource, schemaDraft, type, range (min/max), required flag, consumers[] (repo, path, usage, status), driftStatus (conforming/drifted/missing)

- `metrics[]` — Array of metricEntry objects with: code, name, canonicalSource, schemaDraft, type, range, targetBand, telemetryHooks[], consumers[], driftStatus
- `schemas[]` — Array of schemaArtifact objects with: repo, path, schemaDraft, id, kind, hash, lastModified
- `repos[]` — Array of repoManifestSlice with: name, tier (T1/T2/T3), visibility, role, languages, schemas[], registries[]
- `deadLedger` — Block with zkpSchema, verifierRegistry, totalProofTypes, totalVerifiers, customProofViolations
- `transitions` — ciWorkflows[] and orchestrators[]

VIOLATION V-001

P0

No populated spine-index data file (e.g.,

schema-spine-index-v1.json

) was found in any public repository. The schema is defined but never instantiated. Without a live spine-index, the "single source of truth" claim is aspirational, not operational. All consumer-level drift detection, invariant tracking, and cross-repo schema enumeration is effectively disabled.

2.2 Repo Manifest Schema

LOCATIONS

- `Horror.Place-Orchestrator/schemas/core/hpc-repo-manifest.schema.json`
- `HorrorPlace-Constellation-Contracts/repo-manifest-schema.v1.json`

The manifest schema requires: `manifestVersion`, `repoName`, `tier` (T1/T2/T3), `visibility`, `role`, `languages`, `paths` (schemas/registry/docs/workflows/tooling arrays), `schemas` array (path + schemaRef + kind), `registries` array (path + entityType + format), `ciProfiles`.

VIOLATION V-002

P0

No actual repo manifest JSON files were found for any of the 13 repositories. There is no

hpc-repo-manifest.json

in Horror.Place, Horror.Place-Orchestrator, or HorrorPlace-Constellation-Contracts. The schema exists in two locations but has zero instances.

2.3 Routing Spine

LOCATIONS

- `HorrorPlace-Constellation-Contracts/hpc-routing-spine-v1.json`
- `HorrorPlace-Constellation-Contracts/hpc-routing-spine.schema.json`

The routing spine schema is present and likely defines objectKind-to-repo routing. However, no deterministic routing table was found that maps every `(objectKind, tier)` pair to a unique `(targetRepo, path)`.

2.4 AI Safe Authoring Contract — Partial Routing

LOCATION

`HorrorPlace-Constellation-Contracts/ai-safe-authoring-contract.v1.json`

SCHEMA ID

`https://horror.place/schemas/ai-safe-authoring-contract.v1.json`

This schema comes closest to enforcing deterministic routing. Each artifact in the `artifacts[]` array requires:

- `objectKind` — Enum of 11 values: `dungeonNodeContract`, `dungeonRunContract`, `historySelectorPattern`, `directorPersonaContract`, `directorComfortPolicy`, `moodContract`, `eventContract`, `regionContract`, `styleContract`, `aiChatHorrorProfile`, `registryEntry`
- `targetRepo` — Enum of all 13 repositories
- `path` — Repository-relative file path (validated by regex)
- `tier` — Enum: `Tier1Public`, `Tier2Internal`, `Tier3Vault`, `Tier4Underground`
- `schemaRef` — Canonical schema URI with regex validation

VIOLATION V-003

P1

The authoring contract validates that each artifact declares a targetRepo and tier, but does

NOT

enforce that a given

objectKind+tier

combination maps to exactly one repo. An AI agent could declare

eventContract

targeting

Horror.Place

at

Tier1Public

or

HorrorPlace-Codebase-of-Death

at

Tier2Internal

— both would pass schema validation. There is no

if-then-else

or

$ref

to a routing lookup table. Determinism depends entirely on agent discipline, not schema enforcement.

2.5 Orchestrator Schema-to-Registry Mapping

LOCATION

`Horror.Place-Orchestrator/src/orchestrator.py` — `_schema_to_registry()` method

Current mapping (hardcoded):

Schema Name	Target Registry
`invariants_v1`	`events`
`entertainment_metrics_v1`	`events`
`stylecontract_v1`	`styles`
(all others)	`events` (DEFAULT FALLBACK)

VIOLATION V-004

P0

This mapping is incomplete and non-deterministic:

- Persona schemas → should map to `personas` registry, but fall through to `events`
- Region schemas → should map to `regions` registry, but fall through to `events`
- `moods.json` exists in the registry directory but no schema maps to it
- The default fallback to "events" means any unknown schema silently routes to the wrong registry — a determinism violation

2.6 Registry Path and Format Discrepancy

VIOLATION V-005

P1

Horror.Place README documents registries at

core/registry/*.ndjson

but actual files are at

registry/*.json

. The CircleCI config references

core/registry/*.json

. At least two conflicting path conventions exist. Additionally,

moods.json

exists in the live registry directory but is not documented in the README's

canonical directory structure.

3. Audit Area 2 — Governance Modules

3.1 Module Inventory

Module	Schema Location	H.API Binding	Runtime Code	Status
Consent	`Constellation-Contracts/schemas/consent/` and `consent-state-machine.v1.json`	NONE — no `H.Consent` in `horror_api.lua`	Not found in public repos	SCHEMA ONLY
Age-Tier Router	`Constellation-Contracts/schemas/routing/`	NONE — no `H.Router` in `horror_api.lua`	`consent-state-machine.v1.json` defines ageTier/policyTier routing	SCHEMA ONLY
Budget	NOT FOUND — no `schemas/budget/` directory	NONE	Not found	MISSING
Safety	`Constellation-Contracts/schemas/safety/`	Partial — `safety_adapter.lua` is under `H.Director`, not standalone `H.Safety`	`Orchestrator: lua/H/Director/safety_adapter.lua`	PARTIAL
Templates	NOT FOUND	NONE	Not found	MISSING

Module	Schema Location	H.API Binding	Runtime Code	Status
	— no `schemas/templates/` directory			
Telemetry	`Constellation-Contracts/schemas/telemetry/`	Partial — `H.session_metrics()` stub in `horror_api.lua`	Not found as standalone module	SCHEMA ONLY
Dead-Ledger	`Constellation-Contracts/schemas/security/` (likely); spine-index has `deadLedger` block	NONE — no `H.DeadLedger` in `horror_api.lua`	Signature/DLN validation stubbed in `orchestrator.py`	SCHEMA + STUB

3.2 Consent State Machine — Deep Analysis

LOCATION

`HorrorPlace-Constellation-Contracts/consent-state-machine.v1.json`

SCHEMA ID

`https://horror.place/schemas/consent-state-machine.v1.json`

Well-designed schema defining:

- **States:** Each with `id`, `label`, `ageTier` (minor/adult), `policyTier` (standard/mature/research), and `allowedContentProfile` (6 boolean flags: `allowPsychologicalHorror`, `allowImpliedViolence`, `allowExplicitViolence`, `allowNonSexualNudity`, `allowSexualContent`, `allowSexualViolence`)
- **Transitions:** `from` → `to` state with `allowedForAgeTier`, `requiresExplicitOptIn`, `requiresProof` (proof role array), `cooldownDays`, `requiresHumanReview`, `autoDowngradeOnBreach`

- **Metadata:** `schemaVersion`, `createdAt`, `updatedAt`, `jurisdictionProfile` (global-default, us-18plus, eu-gdpr)

✓ Correct Pattern:

The consent schema properly separates age classification from policy tier and links both to content profiles. The transition system supports ZKP proof requirements and human review gates.

VIOLATION V-006

P1

No

H.Consent

Lua API exists. The

consent-state-machine.v1.json

is not consumed by any public Lua binding. Engine code has no way to call

H.Consent.getState()

,

H.Consent.requestTransition()

, or

H.Consent.checkAllowance()

via the documented API surface. If consent logic exists in Tier-2 repos, it bypasses the public contract layer.

3.3 Safety Adapter — Deep Analysis

LOCATION

`Horror.Place-Orchestrator/lua/H/Director/safety_adapter.lua`

NAMESPACE

`H.Director.SafetyAdapter`

Implements `apply_safety_decision(safetyDecision, metrics, comfortPolicy, boundaryState)` with a strict strategy hierarchy:

```
allow
<
soften
<
imply
<
downshift
<
refuse
```

Checks in priority order:

1. Hard boundary violated → **refuse**
2. DET exceeds absolute cap → **refuse**
3. DET exceeds fraction of session cap → **downshift**
4. CDL exceeds hard upper → **downshift**
5. DET near soft cap → **imply**

6. DET_delta exceeds per-beat cap → **soften**
7. CDL exceeds upper → **soften**

Returns `{strategy, reason}` with concatenated reason strings.

✓ Correct Pattern:

Strategy escalation is monotonic (never weakens), properly consumes DET/CDL/ARR metrics from the published schema, and accepts a comfortPolicy object matching

director-comfort-policy.v1.json

.

VIOLATION V-007

P2

Safety is nested under

H.Director.SafetyAdapter

rather than existing as a standalone

H.Safety

module as the Ten-Module Framework describes. This means safety decisions are only available within the Director's scope — other systems (e.g., PCG seed validation, agent spawning, network layer) cannot call safety checks without going through the Director.

3.4 Missing Modules

VIOLATION V-008

P0

Two governance modules from the Ten-Module Framework are entirely absent:

- **Budget:** No `schemas/budget/` directory in Constellation-Contracts. No budget enforcement logic in any public code. No `H.Budget` API.
- **Templates/Redaction:** No `schemas/templates/` directory. No template enforcement or redaction logic. The authoring contract has `governance.allowedTemplateIds` (a whitelist array), but no template schema or resolver exists to validate those IDs.

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4. Audit Area 3 — Authoring Contracts & Envelope Enforcement

4.1 AI Safe Authoring Contract

LOCATION

`HorrorPlace-Constellation-Contracts/ai-safe-authoring-contract.v1.json`

The contract requires per-session:

- `sessionId` — pattern: `^sess-[a-z0-9]{8,32}$`
- `agentProfileId` — pattern: `^agent-[a-z0-9\-]{3,64}$`
- `aiChatProfileId` — pattern: `^acp-[a-z0-9\-]{3,64}$`

- `mode` — enum: `MapDesignWizard`, `DungeonFlowTuning`, `ContractAuthor`, `TelemetryReview`, `SkeletonFirst`

Each artifact in the `artifacts[]` array declares:

- `objectKind`, `targetRepo`, `path`, `tier`, `schemaRef` (all required)
- `invariants_used`, `metrics_used`, `referencedIds` (optional)

Governance block provides: `requiresDeadLedgerRef` (boolean), `maxIntensityBand` (0–10), `allowedTemplateIds`

VIOLATION V-009

P1

The artifacts array has

minItems: 1

but

NO

maxItems

constraint. The documented "one request → ≤3 artifacts" rule is not enforced at the schema level. An authoring session could declare 50 artifacts and pass validation.

VIOLATION V-010

P1

No "one schema family" constraint exists. The artifacts array allows mixing

objectKinds

> freely — a single session could declare a
>
> dungeonNodeContract
>
> , a
>
> styleContract
>
> , and a
>
> regionContract
>
> simultaneously, violating the "one schema family per request" doctrine.

4.2 File Envelope

LOCATION

`HorrorPlace-Constellation-Contracts/ai_file_envelope_v1.json`

The envelope schema exists and likely wraps authored artifacts with metadata, signatures, and routing information. This follows the prism/envelope pattern.

4.3 Prism Schemas

LOCATIONS IN CONSTELLATION-CONTRACTS

- `schemas/authoring/prism-runtime-binding-v1.json`
- `schemas/authoring/prism-schema-mapping-v1.json`

These schemas define how AI-chat sessions bind to runtime contexts and map to schema families. The presence of both schemas suggests the plan-vs-body narrowing infrastructure is being built at the schema level.

4.4 CHATDIRECTOR

LOCATION

`HorrorPlace-Constellation-Contracts/schemas/chatdirector/`

The CHATDIRECTOR schema directory exists. A Rust binary `hpc-chat-director` is built and used in the Orchestrator's AI authoring validation workflow.

4.5 CI Validation Workflow

LOCATION

`Horror.Place-Orchestrator/.github/workflows/ai-authoring-validate.yml`

This workflow:

8. Triggers on PRs touching `**/*.ai.json`, `**/*ContractCard.v1.json`, or authoring/prism schemas
9. Checks out the Orchestrator repo
10. Attempts to build `hpc-chat-director` from `../HorrorPlace-Constellation-Contracts`
11. Validates AI authoring files by piping through `hpc-chat-director validate-response`

VIOLATION V-011

P2

The workflow references

../HorrorPlace-Constellation-Contracts

as a sibling directory. In standard GitHub Actions, each repository is checked out independently. This path will not resolve unless a custom checkout step fetches both repos. The workflow will fail in CI as currently written.

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5. Audit Area 4 — Telemetry, Safety & Consent Data Flows

5.1 Entertainment Metrics Coverage

Metric	Schema Source	Lua API	Safety Adapter	Spine Index Schema
UEC (User Engagement Coefficient)	`core/schemas/entertainment_metrics_v1.json`	H.session_metrics() stub	Not directly checked	metricEntry with telemetryHooks
EMD (Emotional Modulation Depth)	`core/schemas/entertainment_metrics_v1.json`	H.session_metrics() stub	Not directly checked	metricEntry with telemetryHooks
STCI (Spectral-Temporal Coherence Index)	`core/schemas/entertainment_metrics_v1.json`	H.session_metrics() stub	Not directly checked	metricEntry with telemetryHooks
CDL (Cumulative Distress Level)	`core/schemas/entertainment_metrics_v1.json`	H.session_metrics() stub	YES — hard upper and upper checks	metricEntry with telemetryHooks
ARR (Adaptive Recovery Rate)	`core/schemas/entertainment_metrics_v1.json`	H.session_metrics() stub	Referenced but not directly gated	metricEntry with telemetryHooks
DET (Distress Escalation Threshold)	`core/schemas/invariants_v1.json`	H.DET() convenience accessor	YES — absolute cap, session fraction, soft cap, per-beat delta	invariantEntry with consumers

5.2 Invariants Coverage

All 11 invariants defined in `horror_api.lua`: `CIC`, `MDI`, `AOS`, `RRM`, `FCF`, `SPR`, `RWF`, `DET`, `HVF`, `LSG`, `SHCI` — each with convenience accessor (e.g., `H.CIC()`). The authoring contract's `invariants_used` enum matches: `CIC`, `MDI`, `AOS`, `RRM`, `FCF`, `SPR`, `RWF`, `DET`, `HVF`, `HVF_mag`, `LSG`, `SHCI`.

VIOLATION V-012

P2

HVF_mag

appears in the authoring contract's

invariants_used

enum but not in

horror_api.lua

's

H.invariants()

return table. Either the API surface is missing this field or the authoring contract has an extra entry.

5.3 CIC/DET Caps Enforcement

The `safety_adapter.lua` enforces DET caps through its strategy escalation chain. CIC is listed as an invariant but has no runtime enforcement code in any public repository — it is defined as a "consent invariant contract" conceptual field but lacks a checker.

VIOLATION V-013

P2

CIC enforcement is undefined at runtime. The CIC invariant is the most governance-critical field (it represents the consent-invariant-contract), yet no public code checks or gates on CIC values. The

consent-state-machine.v1.json

defines consent states and transitions but does not reference CIC numerically.

5.4 Missing Invariants Registry

VIOLATION V-014

P2

The architecture document (

HorrorPlace-vm-constellation.md

) describes

H.invariants()

resolving via

registry/invariants.json

, but this file does not exist. The registry directory contains

events.json

,

moods.json

```
,
personas.json
,
regions.json
,
styles.json
— no
invariants.json
. The
H.invariants()
stub in
horror_api.lua
has no backing data store.
```

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6. Wiring Pattern Inventory

6.1 Correct Wiring Patterns Found

Pattern A — Safety Adapter (Schema → Lua API → Runtime)

Flow:

director-comfort-policy.v1.json

schema →

SafetyAdapter.apply_safety_decision()

in Lua → Strategy escalation with DET/CDL metrics → Returns

{strategy, reason}

envelope.

Files:

- **Schema:** `HorrorPlace-Constellation-Contracts/schemas/safety/` and `director-comfort-policy.v1.json`
- **Lua binding:** `Horror.Place-Orchestrator/lua/H/Director/safety_adapter.lua`
- **API surface:** `Horror.Place/horror_api.lua` — `Director.next_beat()` consumes safety decisions

This is the strongest wiring in the constellation — a clean schema-to-code path with monotonic strategy escalation and metric gating.

Pattern B — Artifact Descriptor Pipeline (Schema → CI → Orchestrator)

Flow:

Vault CI produces

artifact_descriptor_v1.json

→ Orchestrator polls →

compute_canonical_hash()

→

verify_canonical_hash()

→

verify_signature()

→

_schema_to_registry()

→ Updates public registry.

Files:

- **Schema:** `Horror.Place-Orchestrator/schemas/artifact_descriptor_v1.json`
- **Orchestrator:** `Horror.Place-Orchestrator/src/orchestrator.py` — `process_descriptor()`
- **Target registries:** `Horror.Place/registry/events.json`, `styles.json`, `personas.json`, `regions.json`
- **CI:** `Horror.Place/.circleci/config.yml` — validates registry entries against schemas

This pipeline is architecturally sound but has gaps in the

_schema_to_registry

mapping and signature verification.

Pattern C — AI Authoring Validation (Schema → Rust Binary → CI)

Flow:

AI agent produces

.ai.json

file → PR triggers workflow →

hpc-chat-director validate-response

checks against

ai-safe-authoring-contract.v1.json

→ Validates objectKind, targetRepo, tier, schemaRef per artifact.

Files:

- **Schema:** `HorrorPlace-Constellation-Contracts/ai-safe-authoring-contract.v1.json`
- **Binary:** `hpc-chat-director` (built from Constellation-Contracts)
- **Workflow:** `Horror.Place-Orchestrator/.github/workflows/ai-authoring-validate.yml`

6.2 Incomplete or Missing Wiring

Module	Schema Exists	Lua API	CI Check	Runtime Enforced	Gap
Spine Index	YES	NO	NO	NO	No instance; no CI generates or validates it
Repo Manifests	YES (×2 locations)	NO	NO	NO	No instances for any repo
Consent	YES	NO	NO	NO	No H.Consent

Module	Schema Exists	Lua API	CI Check	Runtime Enforced	Gap
					API; no runtime consumer
Age-Tier Router	YES (in consent schema)	NO	NO	NO	ageTier/policyTier defined but no router code
Budget	NO	NO	NO	NO	Entirely missing from constellation
Templates	NO	NO	Partial (`allowedTemplateIds` in authoring contract)	NO	No template schemas or resolver
Telemetry	YES	Stub (`H.session_metrics`)	NO	NO	Metrics defined but no collection pipeline
Dead-Ledger	Partial (in spine index and `security/`)	NO	NO	Stub (DLN pattern in descriptor)	Signature verification is stubbed

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7. Violations Summary

ID	Priority	Area	Description	Location	Remediation
V-001	P0	Spine Index	No populated spine-index instance exists	All repos	Generate `schema-spine-index-v1.json` in Constellation-Contracts with all 11 invariants, 5 metrics, 13 repos, and current schema hashes. Add CI job to validate it against `hpc-spine-index.schema.json`.
V-002	P0	Manifests	No repo manifest instances exist for any repository	All repos	Create `hpc-repo-manifest.json` for each of the 13 repos. Add CI validation against `repo-manifest-schema.v1.json`.
V-004	P0	Routing	Orchestrator `_schema_to_registry()` mapping is incomplete; unknown schemas default to "events"	`Orchestrator/src/orchestrator.py`	Add explicit mappings for persona, region, mood schemas. Replace default fallback with a hard error.

ID	Priority	Area	Description	Location	Remediation
V-008	P0	Governance	Budget and Templates modules entirely missing	`Constellation-Contracts`	Create `schemas/budget/` and `schemas/templates/` directories with `budget-envelope.v1.json`, `template-contract.v1.json`, and `template-resolver.v1.json`.
V-003	P1	Routing	Authoring contract does not enforce objectKind +tier → repo determinism	`Constellation-Contracts/ ai-safe-authoring-contract.v1.json`	Add a routing lookup table (or `$ref` to `hpc-routing-spine-v1.json`) with `if-then-else` constraints that enforce the unique mapping.
V-005	P1	Registry	Registry path/format mismatch between docs and actual files	`Horror.Place`	Reconcile README with actual paths. Standardize on `registry/*.json` or `core/registry/*.ndjson`. Update CI config.
V-006	P1	Consent	No `H.Consent` Lua API	`Horror.Place/horror_api.lua`	Add `H.Consent` namespac

ID	Priority	Area	Description	Location	Remediation
			exists		e with `getState()`, `requestTransition()`, `checkAllowance()` stubs. Wire to `consent-state-machine.v1.json`.
V-009	P1	Authoring	No `maxItems:3` on artifacts array	`Constellation-Contracts/ ai-safe-authoring-contract.v1.json`	Add `"maxItems": 3` to the artifacts array definition.
V-010	P1	Authoring	No schema-family constraint on artifacts	`Constellation-Contracts/ ai-safe-authoring-contract.v1.json`	Add a custom validation rule or JSON Schema conditional requiring all artifacts in a session share the same objectKind prefix or schema family.
V-011	P2	CI	Cross-repo sibling path reference in AI authoring workflow	`Orchestrator/.github/workflows/ ai-authoring-validate.yml`	Add explicit checkout step for HorrorPlace-Constellation-Contracts using `actions/ch`

ID	Priority	Area	Description	Location	Remediation
					`eckout` with `path` parameter.
V-007	P2	Safety	Safety nested under `H.Director` instead of standalone `H.Safety`	`Orchestrator/lua/H/Director/safety_adapter.lua`	Promote to `H.Safety` namespace accessible from all systems. Keep `H.Director.SafetyAdapter` as a convenience re-export.
V-012	P2	Metrics	`HVF_mag` in authoring contract but not in `H.invariants()`	`Horror.Place/horror_api.lua` vs `Constellation-Contracts`	Either add `HVF_mag` to `H.invariants()` return table or remove from authoring contract enum.
V-013	P2	CIC	CIC invariant has no runtime enforcement	All public repos	Implement CIC checker in `H.Safety` or `H.Consent` that gates content based on consent state and CIC value.
V-014	P2	Registry	`registry/invariants.json` missing; reference	`Horror.Place/registry/`	Create `registry/invariants.json` with invariant

ID	Priority	Area	Description	Location	Remediation
			d by architecture doc		bundle entries. Update `H.invariants()` resolution path.
V-015	P3	Crypto	KMS signature verification is stubbed	`Orchestrator/src/orchestrator.py`	Implement actual KMS/HSM verify endpoint call. Remove `UNSIGNED-DEV` bypass in production mode.
V-016	P3	Docs	`moods.json` undocumented in README	`Horror.Place/README.md`	Add moods registry to documented directory structure.
V-017	P3	Schema	Two copies of repo manifest schema in different repos	`Orchestrator` + `Contracts`	Consolidate to single canonical source in Constellation-Contracts. Orchestrator should `$ref` it.

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8. Prioritized Remediation Roadmap

Sprint 1 — Foundation (Compiler-Readiness)

1. **Generate populated spine-index instance** — Create `schema-spine-index-v1.json` in Constellation-Contracts with all invariants, metrics, schema hashes, repo manifests, and Dead-Ledger summary. Add CI job.
2. **Create repo manifests for all 13 repos** — One `hpc-repo-manifest.json` per repo, validated against the schema.
3. **Fix `_schema_to_registry()` mapping** — Add persona, region, mood mappings. Replace default fallback with hard error. Add unit tests.
4. **Create Budget and Templates schemas** — `budget-envelope.v1.json`, `template-contract.v1.json`, `template-resolver.v1.json` in Constellation-Contracts.

Sprint 2 — API Surface Completion

1. **Add H.Consent, H.Safety, H.Budget, H.Templates, H.Telemetry, H.DeadLedger** Lua API stubs to `horror_api.lua`, mirroring the existing `H/Director/Persona` pattern.
2. **Promote SafetyAdapter to `H.Safety` namespace** while keeping `H.Director.SafetyAdapter` as re-export.
3. **Implement CIC runtime checker** that gates content based on consent state.
4. **Create `registry/invariants.json`** with invariant bundle entries.

Sprint 3 — Enforcement & Determinism

1. **Add** `maxItems: 3` to authoring contract artifacts array.
2. **Add schema-family constraint** — Either JSON Schema conditional or CHATDIRECTOR binary enforcement.
3. **Build** `objectKind+tier → repo` **routing table** as a standalone JSON file referenced by the authoring contract and CHATDIRECTOR.
4. **Fix AI authoring CI workflow** — Add `actions/checkout` for Constellation-Contracts repo.

Sprint 4 — Production Hardening

1. **Implement KMS signature verification** — Replace stubs with actual HSM/KMS calls.
2. **Reconcile registry paths/formats** — Standardize across README, CI config, and actual files.
3. **Reconcile** `HVF_mag` across authoring contract and `H.invariants()`.
4. **Document** `moods.json` and consolidate repo manifest schema to single source.
5. **Audit private repos** (manual) — Verify Tier-2/T3 repos consume contracts via H.APIs exclusively.

---

9. Cross-Repo Flow Diagrams

9.1 Authoring Flow

```
AI Agent
│
▼ ai-safe-authoring-contract.v1.json  ← Constellation-Contracts
│ (validates objectKind, targetRepo, tier, schemaRef)
▼ hpc-chat-director validate-response  ← Built from Contracts
│
▼ *.ai.json artifact committed to PR  ← Target repo
│
▼ ai-authoring-validate.yml  ← Orchestrator workflow
│ (triggers on *.ai.json, *ContractCard.v1.json)
▼ Schema validation against contracts  ← Constellation-Contracts schemas
```

GAP:

No routing-spine lookup enforces

objectKind+tier

determinism in this flow.

9.2 Vault-to-Registry Pipeline

```
Vault Repo CI (T2/T3)
│ (validates seed/contract against schema)
▼ artifact_descriptor_v1.json  ← Orchestrator schema
│ (source_repo, artifact_type, hash, signature, safetytier)
▼ orchestrator.py: process_descriptor()  ← Orchestrator service
│ 1. Idempotency check
│ 2. Backup existing registry
│ 3. Hash verification (SHA-256)
│ 4. Signature verification
```

```
(STUBBED)
│ 5. _schema_to_registry() mapping
(INCOMPLETE)
│ 6. Build opaque registry entry │ 7. Update target registry │ 8. Audit log ▼ registry/*.json ← Horror.Place core │ ▼ CircleCI: schema-validation job ← Horror.Place CI │ (validates registry entries against core/schemas/) ▼ CircleCI: registry-linter job ← Horror.Place CI │ (NDJSON format, required fields, canonical URIs) ▼ CircleCI: api-leak-test job ← Horror.Place CI (no raw URLs, no base64, no hex binary)
```

GAPS:

Signature verification stubbed.

_schema_to_registry

mapping incomplete.

9.3 Runtime Resolution

```
Engine (Godot/Unreal) │ ▼ H.invariants(region_id, tile_id) ← horror_api.lua │ ▼ Registries.region(region_id) ← registry/regions.json │ ▼ Bundles.load_tile(bundle_ref, tile_id) ← registry/invariants.json
(MISSING)
```

```
│ (hash/signature verification) ▼ Read-only invariant table ← {CIC, MDI, AOS, RRM, FCF, SPR, RWF, DET, HVF, LSG, SHCI} │ ▼ Director.next_beat(session, metrics) ← horror_api.lua │ ▼ SafetyAdapter.apply_safety_decision() ← safety_adapter.lua │ (DET/CDL/ARR gating, strategy escalation) ▼ {strategy, reason} envelope ← allow/soften/imply/downshift/refuse
```

GAP:

registry/invariants.json

does not exist. CIC not gated at runtime.

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10. Conclusion

The Horror$Place VM-constellation demonstrates strong architectural vision: canonical schemas for spine indexing, consent state machines, safe authoring, and safety adaptation are well-designed and follow governance-first principles. The `safety_adapter.lua` is a particularly clean implementation of monotonic strategy escalation against published metric thresholds.

However, the gap between schema design and operational wiring is substantial. The four P0 violations — missing spine-index instance, missing repo manifests, incomplete schema-to-registry routing, and absent Budget/Templates modules — represent foundational infrastructure that must exist before the constellation can be considered "compiler-driven." The system currently operates more as a **"convention-driven"**

architecture where schema compliance depends on developer/agent discipline rather than machine enforcement.

The remediation roadmap is structured to close these gaps progressively: **Sprint 1** establishes the data foundation, **Sprint 2** completes the API surface, **Sprint 3** adds enforcement, and **Sprint 4** hardens for production. Each sprint is scoped for LLM-assisted implementation via Copilot.

Audit scope limitation:

10 of 13 repositories are private and could not be inspected. The most critical verification — whether Tier-2/Tier-3 repos consume contracts exclusively via H.APIs without ad-hoc shortcuts — remains unconfirmed and should be prioritized in a follow-up audit with repository access.

End of Audit Report

Generated: 2026-04-21 08:46 MST | Auditor: Copilot Automated Analysis | Requested by: Doctor0Evil

Next Review: Upon completion of Sprint 1 remediation